niram-css v0.2.0

The `niram-css` package lets you use standard CSS color names directly within your Typst documents.

While Typst includes a curated set of predefined colors, accessing the full spectrum of familiar CSS keywords often requires hunting down exact hex or RGB values. `niram-css` eliminates this friction by bringing all 147 standardized CSS colors natively into your workflow. Instead of pausing to look up a value like `#6495ed`, you can simply type `cornflowerblue` and let the package do the rest.

Key features include:

- **Complete CSS Color Support:** Access all 147 standard color keywords from the CSS3 specification (see Table 1, or view in a browser).
- **Flexible Naming Conventions:** Write color names in the format most natural to you. The package automatically normalizes inputs, meaning `cornflowerblue`, `cornflower blue`, `cornflower-blue`, `cornflower_blue`, and `CornflowerBlue` all resolve to the exact same color.

The word *niram* (நிறம்) means "color" in Tamil, reflecting the package's singular focus on color.

Requirements

This package requires Typst 0.13.0 or higher.

Getting Started

Import the package at the top of your Typst document:

```
#import "@preview/niram-css:0.2.0": *
```

Then, use the `css` function anywhere a color value is expected:

```
Normal text may be #text(fill: css("crimson"))[colored so.]
```

Normal text may be colored so.

That is all it takes to start using CSS color names in your document.

Usage

Function Reference

```
css(color-name: str) -> color
```

color-name `str`

A string representing the color name. The following conventions are all accepted and normalized to a canonical form before lookup:

Style	Example	Common in
No separator	`cornflowerblue`	CSS/SVG
Spaces	`cornflower blue`	X11/GIMP/Matplotlib
Hyphens	`cornflower-blue`	CSS tooling/LaTeX xcolor
Underscores	`cornflower_blue`	X11 alternate/Python/R
CamelCase	`CornflowerBlue`	X11/.NET/Java

Color name matching is completely case-insensitive; `AliceBlue`, `aliceblue`, and `ALICEBLUE` are all treated identically.

-> `color`

The `css` function returns an RGB color. Because this returned value is a standard Typst color type, you can seamlessly modify it using any of Typst's built-in color functions, such as `.lighten()` or `.darken()`.

Examples

All the examples below assume `#import "@preview/niram-css:0.2.0": *` is at the top of the document.

1. Filling a rectangle with a CSS color.

```
#rect(width: 100%, height: auto, fill: css("darkorchid"))
```

2. CSS color names can be written in lowercase or CamelCase. Both variants are valid.

```
(a) saddlebrown: #box(width: 30mm, height: 1em, fill: css("saddlebrown"))

(b) SaddleBrown: #box(width: 30mm, height: 1em, fill: css("SaddleBrown"))
```

(a) saddlebrown:

(b) SaddleBrown:

3. Typst's built-in `maroon` has a different hex value than the standard CSS `maroon`. To highlight this distinction, the CSS version in the example below uses an initial uppercase letter (`Maroon`).

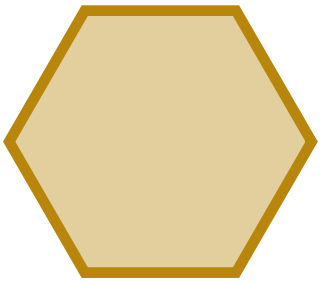
```
#grid(
  columns: 3,
  rows: 2,
```

```
  row-gutter: 10pt,
  column-gutter: 10pt,
  [Typst maroon], [#color.maroon.to-hex()], [#box(width: 30mm, height: 1em,
fill: maroon)],
  [CSS maroon], [#css("Maroon").to-hex()], [#box(width: 30mm, height: 1em,
fill: css("Maroon"))]
)
```

Typst maroon	#85144b	
CSS maroon	#800000	

4. Using CSS colors in drawing and data visualization.

```
#align(center)[
  #polygon.regular(
    fill: css("darkgoldenrod").lighten(60%),
    stroke: (paint: css("darkgoldenrod"), thickness: 4pt, cap: "round"),
    size: 60mm,
    vertices: 6,
  )
]
```



5. Using CSS colors to underline text.

```
This is #underline(stroke: css("teal"))[important.]

This is #underline(stroke: (paint: css("teal"), thickness: 2pt))[important.]
```

This is important.

This is important.

Background: The CSS Color Standard

X11 Colors

The X11 color names [1], developed in the 1980s, included a file named `rgb.txt` that was shipped with every X11 installation. It mapped over 400 color names to specific RGB values. This list was used in Unix-based systems to standardize color representation, but the list was never formally standardized across web formats, leading to variations between distributions.

Sixteen Named Colors

The HTML 3.2 specification [2] of 1996 introduced sixteen color names, derived from the standard 16 colors supported by the Windows VGA palette. These colors, called the sixteen named colors, are:

- `aqua`
- `black`
- `blue`
- `fuchsia`
- `gray`
- `green`
- `lime`
- `maroon`
- `navy`
- `olive`
- `purple`
- `red`
- `silver`
- `teal`
- `white`
- `yellow`

Thanks to this standard, these sixteen color names could be used reliably and consistently in documents across the Web, without needing to specify hexadecimal values.

The CSS Level 1 colors introduced in 1996 specified these exact same sixteen colors, while the CSS Level 2 colors, introduced in 1998, expanded to seventeen colors with the addition of `orange`.

CSS/SVG Colors

Finally, the SVG 1.1 Specification of 2011 [3] and the CSS Color Module Level 3 [4] of 2022 standardized on a list of 147 color names, known as the "recognized color keyword names." They comprise the original HTML/VGA named colors, `orange`, and additional colors drawn from the X11 set.

This standardized list provides a consistent color naming scheme across the Web and has remained stable since its adoption. It is this set of color names that forms the basis for this package. How the CSS colors were given their rather idiosyncratic names is explained in a lighthearted but factual fashion by Alex Sexton in this YouTube video [5].

While all 147 CSS color names are distinct, only 138 have unique hex values. The nine pairs of color names that share identical hex values are detailed in Table 2.

CSS Colors vs. Typst Built-ins

Typst ships with eighteen predefined colors, including the sixteen HTML named colors, `orange`, and one additional color named `eastern`. The color `eastern` has no CSS equivalent.

The remaining seventeen Typst colors share names with their CSS counterparts, but—with the exception of white and black—they have different hex values, as shown in Table 3.

X11 Gray Equivalents

X11 color coverage is intentionally out of scope for this package. The X11 database includes numbered color families (e.g. `red1–red4`, `blue1–blue4`, `gray0–gray100`) that are rarely used today.

However, any X11 numbered gray shade can be reproduced exactly without this package using `black.lighten(n%)`, which produces the same result as the corresponding X11 gray. These equivalents are detailed in Table 4.

CSS Color Reference

Color Name	Hex Value	Swatch
aliceblue	`#f0f8ff`	
antiquewhite	`#faebd7`	
aqua	`#00ffff`	
aquamarine	`#7fffd4`	
azure	`#f0ffff`	
beige	`#f5f5dc`	
bisque	`#ffe4c4`	
black	`#000000`	
blanchedalmond	`#ffebcd`	
blue	`#0000ff`	
blueviolet	`#8a2be2`	
brown	`#a52a2a`	
burlywood	`#deb887`	

Color Name	Hex Value	Swatch
cadetblue	`#5f9ea0`	
chartreuse	`#7fff00`	
chocolate	`#d2691e`	
coral	`#ff7f50`	
cornflowerblue	`#6495ed`	
cornsilk	`#fff8dc`	
crimson	`#dc143c`	
cyan	`#00ffff`	
darkblue	`#00008b`	
darkcyan	`#008b8b`	
darkgoldenrod	`#b8860b`	
darkgray	`#a9a9a9`	
darkgreen	`#006400`	

Color Name	Hex Value	Swatch
darkgrey	`#a9a9a9`	
darkkhaki	`#bdb76b`	
darkmagenta	`#8b008b`	
darkolivegreen	`#556b2f`	
darkorange	`#ff8c00`	
darkorchid	`#9932cc`	
darkred	`#8b0000`	
darksalmon	`#e9967a`	
darkseagreen	`#8fbc8f`	
darkslateblue	`#483d8b`	
darkslategray	`#2f4f4f`	
darkslategrey	`#2f4f4f`	
darkturquoise	`#00ced1`	

Color Name	Hex Value	Swatch
darkviolet	`#9400d3`	
deeppink	`#ff1493`	
deepskyblue	`#00bfff`	
dimgray	`#696969`	
dimgrey	`#696969`	
dodgerblue	`#1e90ff`	
firebrick	`#b22222`	
floralwhite	`#fffaf0`	
forestgreen	`#228b22`	
fuchsia	`#ff00ff`	
gainsboro	`#dcdcdc`	
ghostwhite	`#f8f8ff`	
gold	`#ffd700`	

Color Name	Hex Value	Swatch
goldenrod	`#daa520`	
gray	`#808080`	
green	`#008000`	
greenyellow	`#adff2f`	
grey	`#808080`	
honeydew	`#f0fff0`	
hotpink	`#ff69b4`	
indianred	`#cd5c5c`	
indigo	`#4b0082`	
ivory	`#fffff0`	
khaki	`#f0e68c`	
lavender	`#e6e6fa`	
lavenderblush	`#fff0f5`	

Color Name	Hex Value	Swatch
lawngreen	`#7cfc00`	
lemonchiffon	`#fffacd`	
lightblue	`#add8e6`	
lightcoral	`#f08080`	
lightcyan	`#e0ffff`	
lightgoldenrodyellow	`#fafad2`	
lightgray	`#d3d3d3`	
lightgreen	`#90ee90`	
lightgrey	`#d3d3d3`	
lightpink	`#ffb6c1`	
lightsalmon	`#ffa07a`	
lightseagreen	`#20b2aa`	
lightskyblue	`#87cefa`	

Color Name	Hex Value	Swatch
lightslategray	#778899	
lightslategrey	#778899	
lightsteelblue	#b0c4de	
lightyellow	#ffffe0	
lime	#00ff00	
limegreen	#32cd32	
linen	#faf0e6	
magenta	#ff00ff	
maroon	#800000	
mediumaquamarine	#66cdaa	
mediumblue	#0000cd	
mediumorchid	#ba55d3	
mediumpurple	#9370db	

Color Name	Hex Value	Swatch
mediumseagreen	#3cb371	
mediumslateblue	#7b68ee	
mediumspringgreen	#00fa9a	
mediumturquoise	#48d1cc	
mediumvioletred	#c71585	
midnightblue	#191970	
mintcream	#f5fffa	
mistyrose	#ffe4e1	
moccasin	#ffe4b5	
navajowhite	#ffdead	
navy	#000080	
oldlace	#fdf5e6	
olive	#808000	

Color Name	Hex Value	Swatch
olivedrab	`#6b8e23`	
orange	`#ffa500`	
orangered	`#ff4500`	
orchid	`#da70d6`	
palegoldenrod	`#eee8aa`	
palegreen	`#98fb98`	
paleturquoise	`#afeeee`	
palevioletred	`#db7093`	
papayawhip	`#ffefd5`	
peachpuff	`#ffdab9`	
peru	`#cd853f`	
pink	`#ffc0cb`	
plum	`#dda0dd`	

Color Name	Hex Value	Swatch
powderblue	`#b0e0e6`	
purple	`#800080`	
red	`#ff0000`	
rosybrown	`#bc8f8f`	
royalblue	`#4169e1`	
saddlebrown	`#8b4513`	
salmon	`#fa8072`	
sandybrown	`#f4a460`	
seagreen	`#2e8b57`	
seashell	`#fff5ee`	
sienna	`#a0522d`	
silver	`#c0c0c0`	
skyblue	`#87ceeb`	

Color Name	Hex Value	Swatch
slateblue	`#6a5acd`	
slategray	`#708090`	
slategrey	`#708090`	
snow	`#fffafa`	
springgreen	`#00ff7f`	
steelblue	`#4682b4`	
tan	`#d2b48c`	
teal	`#008080`	
thistle	`#d8bfd8`	
tomato	`#ff6347`	
turquoise	`#40e0d0`	
violet	`#ee82ee`	
wheat	`#f5deb3`	

Color Name	Hex Value	Swatch
white	`#ffffff`	
whitesmoke	`#f5f5f5`	
yellow	`#ffff00`	
yellowgreen	`#9acd32`	

Table 1: All 147 CSS named colors with their hex values and swatches.

CSS Color Aliases

Color Names	Hex Value	Swatch
aqua cyan	`#00ffff`	
darkgray darkgrey	`#a9a9a9`	
darkslategray darkslategrey	`#2f4f4f`	
dimgray dimgrey	`#696969`	
fuchsia magenta	`#ff00ff`	
gray grey	`#808080`	
lightgray lightgrey	`#d3d3d3`	
lightslategray lightslategrey	`#778899`	
slategray slategrey	`#708090`	

Table 2: CSS colors sharing the same hex value.

Typst vs. CSS Color Values

Color Name	Typst Hex Value	Swatch	CSS Hex Value	Swatch
aqua	`#7fdbff`		`#00ffff`	
black	`#000000`		`#000000`	
blue	`#0074d9`		`#0000ff`	
eastern	`#239dad`		—	—
fuchsia	`#f012be`		`#ff00ff`	
gray	`#aaaaaa`		`#808080`	
green	`#2ecc40`		`#008000`	
lime	`#01ff70`		`#00ff00`	
maroon	`#85144b`		`#800000`	
navy	`#001f3f`		`#000080`	
olive	`#3d9970`		`#808000`	
orange	`#ff851b`		`#ffa500`	
purple	`#b10dc9`		`#800080`	

Color Name	Typst Hex Value	Swatch	CSS Hex Value	Swatch
red	`#ff4136`		`#ff0000`	
silver	`#dddddd`		`#c0c0c0`	
teal	`#39cccc`		`#008080`	
white	`#ffffff`		`#ffffff`	
yellow	`#ffdc00`		`#ffff00`	

Table 3: Hex value differences between Typst's predefined colors and their CSS equivalents.

X11 Gray Equivalents

X11 Name	Typst Expression	Hex Value	Swatch
gray0 grey0	black.lighten(0%)	#000000	
gray1 grey1	black.lighten(1%)	#030303	
gray2 grey2	black.lighten(2%)	#050505	
gray3 grey3	black.lighten(3%)	#080808	
gray4 grey4	black.lighten(4%)	#0a0a0a	
gray5 grey5	black.lighten(5%)	#0d0d0d	
gray6 grey6	black.lighten(6%)	#0f0f0f	
gray7 grey7	black.lighten(7%)	#121212	
gray8 grey8	black.lighten(8%)	#141414	
gray9 grey9	black.lighten(9%)	#171717	
gray10 grey10	black.lighten(10%)	#1a1a1a	
gray11 grey11	black.lighten(11%)	#1c1c1c	
gray12 grey12	black.lighten(12%)	#1f1f1f	

X11 Name	Typst Expression	Hex Value	Swatch
gray13 grey13	black.lighten(13%)	#212121	
gray14 grey14	black.lighten(14%)	#242424	
gray15 grey15	black.lighten(15%)	#262626	
gray16 grey16	black.lighten(16%)	#292929	
gray17 grey17	black.lighten(17%)	#2b2b2b	
gray18 grey18	black.lighten(18%)	#2e2e2e	
gray19 grey19	black.lighten(19%)	#303030	
gray20 grey20	black.lighten(20%)	#333333	
gray21 grey21	black.lighten(21%)	#363636	
gray22 grey22	black.lighten(22%)	#383838	
gray23 grey23	black.lighten(23%)	#3b3b3b	
gray24 grey24	black.lighten(24%)	#3d3d3d	
gray25 grey25	black.lighten(25%)	#404040	

X11 Name	Typst Expression	Hex Value	Swatch
gray26 grey26	black.lighten(26%)	#424242	
gray27 grey27	black.lighten(27%)	#454545	
gray28 grey28	black.lighten(28%)	#474747	
gray29 grey29	black.lighten(29%)	#4a4a4a	
gray30 grey30	black.lighten(30%)	#4c4c4c	
gray31 grey31	black.lighten(31%)	#4f4f4f	
gray32 grey32	black.lighten(32%)	#525252	
gray33 grey33	black.lighten(33%)	#545454	
gray34 grey34	black.lighten(34%)	#575757	
gray35 grey35	black.lighten(35%)	#595959	
gray36 grey36	black.lighten(36%)	#5c5c5c	
gray37 grey37	black.lighten(37%)	#5e5e5e	
gray38 grey38	black.lighten(38%)	#616161	

X11 Name	Typst Expression	Hex Value	Swatch
gray39 grey39	black.lighten(39%)	#636363	
gray40 grey40	black.lighten(40%)	#666666	
gray41 grey41	black.lighten(41%)	#696969	
gray42 grey42	black.lighten(42%)	#6b6b6b	
gray43 grey43	black.lighten(43%)	#6e6e6e	
gray44 grey44	black.lighten(44%)	#707070	
gray45 grey45	black.lighten(45%)	#737373	
gray46 grey46	black.lighten(46%)	#757575	
gray47 grey47	black.lighten(47%)	#787878	
gray48 grey48	black.lighten(48%)	#7a7a7a	
gray49 grey49	black.lighten(49%)	#7d7d7d	
gray50 grey50	black.lighten(50%)	#808080	
gray51 grey51	black.lighten(51%)	#828282	

X11 Name	Typst Expression	Hex Value	Swatch
gray52 grey52	black.lighten(52%)	#858585	
gray53 grey53	black.lighten(53%)	#878787	
gray54 grey54	black.lighten(54%)	#8a8a8a	
gray55 grey55	black.lighten(55%)	#8c8c8c	
gray56 grey56	black.lighten(56%)	#8f8f8f	
gray57 grey57	black.lighten(57%)	#919191	
gray58 grey58	black.lighten(58%)	#949494	
gray59 grey59	black.lighten(59%)	#969696	
gray60 grey60	black.lighten(60%)	#999999	
gray61 grey61	black.lighten(61%)	#9c9c9c	
gray62 grey62	black.lighten(62%)	#9e9e9e	
gray63 grey63	black.lighten(63%)	#a1a1a1	
gray64 grey64	black.lighten(64%)	#a3a3a3	

X11 Name	Typst Expression	Hex Value	Swatch
gray65 grey65	black.lighten(65%)	#a6a6a6	
gray66 grey66	black.lighten(66%)	#a8a8a8	
gray67 grey67	black.lighten(67%)	#ababab	
gray68 grey68	black.lighten(68%)	#adadad	
gray69 grey69	black.lighten(69%)	#b0b0b0	
gray70 grey70	black.lighten(70%)	#b2b2b2	
gray71 grey71	black.lighten(71%)	#b5b5b5	
gray72 grey72	black.lighten(72%)	#b8b8b8	
gray73 grey73	black.lighten(73%)	#bababa	
gray74 grey74	black.lighten(74%)	#bdbdbd	
gray75 grey75	black.lighten(75%)	#bfbfbf	
gray76 grey76	black.lighten(76%)	#c2c2c2	
gray77 grey77	black.lighten(77%)	#c4c4c4	

X11 Name	Typst Expression	Hex Value	Swatch
gray78 grey78	black.lighten(78%)	#c7c7c7	
gray79 grey79	black.lighten(79%)	#c9c9c9	
gray80 grey80	black.lighten(80%)	#cccccc	
gray81 grey81	black.lighten(81%)	#cfcfcf	
gray82 grey82	black.lighten(82%)	#d1d1d1	
gray83 grey83	black.lighten(83%)	#d4d4d4	
gray84 grey84	black.lighten(84%)	#d6d6d6	
gray85 grey85	black.lighten(85%)	#d9d9d9	
gray86 grey86	black.lighten(86%)	#dbdbdb	
gray87 grey87	black.lighten(87%)	#dedede	
gray88 grey88	black.lighten(88%)	#e0e0e0	
gray89 grey89	black.lighten(89%)	#e3e3e3	
gray90 grey90	black.lighten(90%)	#e6e6e6	

X11 Name	Typst Expression	Hex Value	Swatch
gray91 grey91	black.lighten(91%)	#e8e8e8	
gray92 grey92	black.lighten(92%)	#ebebeb	
gray93 grey93	black.lighten(93%)	#ededed	
gray94 grey94	black.lighten(94%)	#f0f0f0	
gray95 grey95	black.lighten(95%)	#f2f2f2	
gray96 grey96	black.lighten(96%)	#f5f5f5	
gray97 grey97	black.lighten(97%)	#f7f7f7	
gray98 grey98	black.lighten(98%)	#fafafa	
gray99 grey99	black.lighten(99%)	#fcfcfc	
gray100 grey100	black.lighten(100%)	#ffffff	

Table 4: X11 numbered gray equivalents using `black.lighten(n%)`.

References

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